

State feedback $\mathcal{H}_\infty$ control design of continuous-time switched affine systems

Grace S. Deaecto , Guilherme C. Santos

DMC/School of Mechanical Engineering, UNICAMP 13083-860, Campinas, SP, Brazil

✉ E-mail: grace@fem.unicamp.br

ISSN 1751-8644

Received on 12th February 2014

Revised on 26th October 2014

Accepted on 26th November 2014

doi: 10.1049/iet-cta.2014.0153

www.ietdl.org

Abstract: This study deals with state feedback $\mathcal{H}_\infty$ control design of continuous-time switched affine systems. The main purpose is to design a set of state feedback gains together with a switching function assuring global asymptotic stability of a desired equilibrium point. The equilibrium point belongs to a set of attainable ones to be determined. Moreover, the control design must take into account a pre-specified upper bound to the $\mathcal{L}_2$ gain from the external input to the controlled output. Two different switching functions are proposed and discussed. The first one depends only on the state and the other depends also on the external input. The results are compared with recent ones available in the literature to date, as for instance, those based on a max-type Lyapunov function and those commonly used to assure practical stability. Numerical examples illustrate the theoretical results and are used for comparisons.

1 Introduction

Switched systems have attracted the attention of the scientific community in the last decades because of their high potential for practical applications, as for instance, in power electronics [1–3]. They represent a subclass of hybrid systems characterised by having a switching rule that selects at each instant of time one of the available subsystems. This switching rule can be arbitrary, see [4], or a control variable that must be designed in order to assure stability and good performance for the overall system. Another possible control technique is based on sliding mode control, as it has been proposed in [5]. For switched linear systems, the literature presents several important results concerning the control design of a switching rule based on state feedback [6, 7] and on output feedback [8, 9]. One interesting characteristic is that the switching strategy can assure global asymptotic stability even if all subsystems are unstable. Moreover, the recent paper [10] shows that, even if all subsystems are stable, a consistent switching rule can enhance the global performance when compared with that of each isolated subsystem. The articles [11–13] and the books [14, 15] are surveys on this topic.

Switched affine systems are more involved since they present several equilibrium points composing a region on the state space. Hence, the control goals consist in determining a set of attainable equilibrium points and a switching rule that guides any trajectory of the system, starting from an arbitrary initial condition, to the desired equilibrium, which can not be shared by any of the subsystems. The literature presents some results about stabilisation based on optimal control, see [2, 16, 17], and on Lyapunov method approach, see [1, 3, 18–25], where [19, 25] deal with sampled-data implementation. Most of them, as for instance, [3, 18] are based on a quadratic Lyapunov function and show that the existence of a stable convex combination of the subsystems state-space matrices is a sufficient condition for global asymptotic stability. Another recent technique is based on a max-type Lyapunov function, see [21–24]. Unfortunately, using this technique, the sufficient condition just mentioned becomes only necessary. Regarding $\mathcal{H}_\infty$ performance, to the best of the authors' knowledge, the literature presents very few results, see [24] as an example.

This paper generalises the results of [3] to cope with $\mathcal{H}_\infty$ control design of switched affine systems. As a first step, we treat the stabilisation problem taking into account the design of a stabilising switching function as the unique control variable. Two different types of switching functions are considered. The first depends on

the state and the other, more general, is a novel proposal that depends also on the external input. As a second step, we generalise these results to cope with the joint design of a set of state feedback gains and a switching function assuring an $\mathcal{H}_\infty$ performance. In this case, because of the action of the two control variables, it is possible to assure global asymptotic stability even if there does not exist a stable convex combination of the subsystems state space matrices. This point will be illustrated in Example 2. As it will be clear in the paper, for special subclasses of switched affine systems, both techniques admit a representative transfer function. The results on stabilisation are theoretically compared with recent works available in the literature, as for instance, those based on a max-type Lyapunov function and those commonly used to assure practical stability. Compared to these works, a great advantage is that the proposed technique can cope with more general control design problems, as the state feedback control design that will be presented afterwards. The results are illustrated by numerical examples.

The notation is standard. For real matrices or vectors ($'$) indicates transpose. For symmetric matrices, the symbol ($\bullet$) denotes each of its symmetric blocks and I denotes the identity matrix. The convex combination of matrices with the same dimension $\{J_1, \dots, J_N\}$ is denoted by $J_\lambda = \sum_{j=1}^N \lambda_j J_j$, where λ belongs to the unitary simplex Λ composed by all non-negative vectors $\lambda \in \mathbb{R}^N$ such that $\sum_{j=1}^N \lambda_j = 1$. The squared norm of a trajectory $\xi(t)$ defined for all $t \geq 0$, denoted by $\|\xi\|_2^2$, is equal to $\|\xi\|_2^2 = \int_0^\infty \xi(t)' \xi(t) dt$. The set of all finite norm trajectories, such that $\|\xi\|_2^2 < \infty$ is denoted by $\mathcal{L}_2$. Finally, the set composed by the first N positive integers, namely $\{1, \dots, N\}$, is denoted by $\mathbb{K}$.

2 Problem statement

Consider a continuous-time switched affine system with the state-space realisation

$$\dot{x}(t) = A_\sigma x(t) + B_\sigma u(t) + H_\sigma w(t) + b_\sigma, \quad x(0) = x_e \quad (1)$$

$$z(t) = E_\sigma x(t) + F_\sigma u(t) + G_\sigma w(t) \quad (2)$$

where $x(t) \in \mathbb{R}^{n_x}$ is the state, $u(t) \in \mathbb{R}^{n_u}$ is the control input, $w(t) \in \mathbb{R}^{n_w}$ is the external input, supposed to belong to $\mathcal{L}_2$ and $z(t) \in \mathbb{R}^{n_z}$ is the controlled output. The switching function $\sigma(t) : t \geq 0 \rightarrow \mathbb{K}$

selects at each instant of time $t \geq 0$ one of the N available affine subsystems. Owing to the affine term $b_i \neq 0$, for at least one $i \in \mathbb{K}$, systems (1)–(2) has several equilibrium points defining a region denoted by $X_e \subset \mathbb{R}^{n_x}$. These points can be attained by a suitable switching policy, see [3]. Clearly, whenever $b_i = 0$ for all $i \in \mathbb{K}$, system (1)–(2) becomes linear and the unique equilibrium point is the origin.

In this paper, our main goal is to determine a set of state feedback gains $\{K_1, \dots, K_N\}$ to compose the control law

$$u(t) = K_\sigma(x(t) - x_e) \quad (3)$$

together with a closed-loop switching strategy $\sigma(x) : \mathbb{R}^{n_x} \rightarrow \mathbb{K}$ in order to guide the system trajectories starting from any initial condition $x_0 \in \mathbb{R}^{n_x}$ to a desired equilibrium point $x_e \in X_e \subset \mathbb{R}^{n_x}$, where $X_e \subset \mathbb{R}^{n_x}$ is a set to be determined. Moreover, the control design must take into account an $\mathcal{H}_\infty$ performance cost defined in the sequel. Actually, our purpose is to generalise the results of [3] to cope with $\mathcal{H}_\infty$ control design and go further by proposing a novel and less conservative state-input dependent switching function $\sigma(x, w) : \mathbb{R}^{n_x} \times \mathbb{R}^{n_w} \rightarrow \mathbb{K}$. Defining the vector $\xi(t) = x(t) - x_e$, the system becomes

$$\dot{\xi}(t) = A_\sigma \xi(t) + B_\sigma u(t) + H_\sigma w(t) + \ell_\sigma, \quad \xi(0) = 0 \quad (4)$$

$$z_e(t) = E_\sigma \xi(t) + F_\sigma u(t) + G_\sigma w(t) \quad (5)$$

where $\ell_\sigma = A_\sigma x_e + b_\sigma$ is the affine term and $z_e(t) = z(t) - E_\sigma x_e$ is the controlled output shifted to the equilibrium point x_e .

Considering systems (1)–(2) with $u(t) = 0$, $\forall t \geq 0$, the control goals are accomplished whenever the switching function $\sigma(\xi(t))$ guides the trajectory $\xi(t)$ to the origin assuring that the $\mathcal{H}_\infty$ performance index satisfies

$$J_\infty(\sigma) = \sup_{0 \neq w \in \mathcal{L}_2} \frac{\|z_e\|_2^2}{\|w\|_2^2} < \rho \quad (6)$$

for some $\rho > 0$ given. The rationale behind this definition is that, whenever A_i is Hurwitz and the switching function is fixed, that is, $\sigma(t) = i$ for all $t \geq 0$, this index equals the standard $\mathcal{H}_\infty$ squared norm of the i th subsystem transfer function from the input w to the output z_e . Indeed, notice that for a fixed switching function $\sigma(t) = i$, $t \geq 0$, the only equilibrium point is $x_e = -A_i^{-1}b_i$ and the system becomes linear, that is

$$\dot{\xi}(t) = A_i \xi(t) + H_i w(t), \quad \xi(0) = 0 \quad (7)$$

$$z_e(t) = E_i \xi(t) + G_i w(t) \quad (8)$$

whose $\mathcal{H}_\infty$ squared norm is well defined, being given in (6). However, notice that system (4)–(5) depends non-linearly on $\sigma(\cdot)$ and, therefore the $\mathcal{H}_\infty$ performance index (6) becomes virtually impossible to calculate. Hence, the idea is to design a switching function such that the upper bound $\rho > 0$ in (6) is minimised.

Connecting the control law (3) to systems (4)–(5), we obtain

$$\dot{\xi}(t) = (A_\sigma + B_\sigma K_\sigma) \xi(t) + H_\sigma w(t) + \ell_\sigma, \quad \xi(0) = 0 \quad (9)$$

$$z_e(t) = (E_\sigma + F_\sigma K_\sigma) \xi(t) + G_\sigma w(t) \quad (10)$$

It is interesting to notice that the affine term ℓ_σ , that will be used to obtain the set of attainable equilibrium points, does not depend on the state feedback gains K_i , $i \in \mathbb{K}$. The gains are only present on the linear part and allow to assure global asymptotic stability even if there does not exist a Hurwitz convex combination of matrices A_i , $i \in \mathbb{K}$.

3 State feedback control design

In this section, it will be presented two different switching functions for the $\mathcal{H}_\infty$ control design of the switched affine system

(1)–(2). Both are based on a quadratic Lyapunov function

$$v(\xi) = \xi' P \xi \quad (11)$$

with $P > 0$. As already mentioned, the first switching rule is state dependent and is a generalisation of the results of [3] to cope with $\mathcal{H}_\infty$ control. The other rule depends also on the external input $w \in \mathcal{L}_2$ and provides less conservative results. Before beginning, let us define the following matrix functions

$$\mathcal{L}_i(\rho, P) = \begin{bmatrix} A_i' P + P A_i & \bullet \\ H_i' P & -\rho I \end{bmatrix} + \begin{bmatrix} E_i' \\ G_i' \end{bmatrix} \begin{bmatrix} E_i' \\ G_i' \end{bmatrix}' \quad (12)$$

$$\mathcal{N}_i(\rho, P) = A_i' P + P A_i + E_i' E_i + (H_i' P + G_i' E_i)' (\rho I - G_i' G_i)^{-1} (H_i' P + G_i' E_i) \quad (13)$$

for $i \in \mathbb{K}$, which will be largely used afterwards.

3.1 State dependent switching function

Next theorem considers systems (1)–(2) with $u(t) = 0$, $\forall t \geq 0$, and provides a state dependent switching function assuring that inequality (6) is verified for $\rho > 0$ given.

Theorem 1: Consider system (1)–(2) with $u(t) = 0$, $\forall t \geq 0$, and let $x_e \in \mathbb{R}^{n_x}$ be given. If there exist $\lambda \in \Lambda$, a positive definite matrix $P \in \mathbb{R}^{n_x \times n_x}$, symmetric matrices $Q_i \in \mathbb{R}^{n_x \times n_x}$ and a scalar $\rho > 0$ satisfying the conditions

$$\begin{bmatrix} A_i' P + P A_i - Q_i & \bullet & \bullet \\ H_i' P & -\rho I & \bullet \\ E_i & G_i & -I \end{bmatrix} < 0, \quad i \in \mathbb{K} \quad (14)$$

$$A_\lambda x_e + b_\lambda = 0 \quad (15)$$

$$Q_\lambda < 0 \quad (16)$$

then the state dependent switching strategy

$$\sigma(x) = \arg \min_{i \in \mathbb{K}} (x - x_e)' (Q_i (x - x_e) + 2P \ell_i) \quad (17)$$

makes the equilibrium point $x_e \in \mathbb{R}^{n_x}$ globally asymptotically stable and assures that (6) holds.

Proof: Consider that conditions (14)–(16) are satisfied and adopt the switching function (17). The time derivative of (11) along an arbitrary trajectory of (4)–(5) provides

$$\begin{aligned} \dot{v}(\xi) &= \dot{\xi}' P \xi + \xi' P \dot{\xi} \\ &= (2\xi' P (A_\sigma \xi + H_\sigma w + \ell_\sigma) + z_e' z_e - \rho w' w) - z_e' z_e + \rho w' w \\ &= \begin{bmatrix} \xi \\ w \end{bmatrix}' \mathcal{L}_\sigma(\rho, P) \begin{bmatrix} \xi \\ w \end{bmatrix} + 2\xi' P \ell_\sigma - z_e' z_e + \rho w' w \\ &\leq \xi' \mathcal{N}_\sigma(\rho, P) \xi + 2\xi' P \ell_\sigma - z_e' z_e + \rho w' w \\ &< \min_{i \in \mathbb{K}} \xi' (Q_i \xi + 2P \ell_i) - z_e' z_e + \rho w' w \\ &< \min_{\lambda \in \Lambda} \xi' (Q_\lambda \xi + 2P \ell_\lambda) - z_e' z_e + \rho w' w \\ &< -z_e' z_e + \rho w' w \end{aligned} \quad (18)$$

where the first inequality comes from the identity

$$\sup_{w \in \mathcal{L}_2} \begin{bmatrix} \xi \\ w \end{bmatrix}' \mathcal{L}_\sigma(\rho, P) \begin{bmatrix} \xi \\ w \end{bmatrix} = \xi' \mathcal{N}_\sigma(\rho, P) \xi \quad (19)$$

Moreover, performing successively the Schur complement in (14) with respect to the third and second rows and columns, we have $\mathcal{N}_i(\rho, P) < Q_i, \forall i \in \mathbb{K}$. Consequently, the second and third inequalities of (18) follow from the adoption of the switching rule (17). Finally, the fourth inequality comes directly from (15) remembering that $\ell_\lambda = A_\lambda x_e + b_\lambda = 0$ and (16) holds. Notice that, setting $w = 0$ for all $t \geq 0$, we have $\dot{v}(\xi) < 0$ and, therefore, the equilibrium point x_e is globally asymptotically stable. Then, for $0 \neq w \in \mathcal{L}_2$, integrating (18) both sides from $t = 0$ to $t \rightarrow \infty$, since $v(\xi(0)) = 0$ and $v(\xi(\infty)) = 0$, we have $J_\infty(\sigma) < \rho$ and the proof is concluded. $\square$

A possibly easier manner of solving the conditions of Theorem 1 is to first determine the set of attainable equilibrium points by satisfying (15) for all $\lambda \in \Lambda$. Indeed, notice that the first diagonal block of inequalities (14) together with (16) assure that A_λ is Hurwitz. We denote the set of all Hurwitz matrices A_λ , $\lambda \in \Lambda$ by $\mathcal{H}$. This implies that all equilibrium points satisfying (15) belong to the set

$$X_e := \{-A_\lambda^{-1}b_\lambda : A_\lambda \in \mathcal{H}, \forall \lambda \in \Lambda\} \quad (20)$$

which can be numerically determined. Only equilibrium points belonging to $X_e \subset \mathbb{R}^{n_x}$ can be attained by the switching function (17). Hence, for each $x_e \in X_e$ there exists an associated vector $\lambda(x_e) \in \Lambda$ which is used in (14) to obtain P , important to implement the switching function (17). Actually, matrix $P > 0$ depends on λ and needs to be recalculated for each different choice of $x_e \in X_e$. An interesting point of Theorem 1 is that the affine part ℓ_i , $i \in \mathbb{K}$, and the linear part $(A_i, H_i, E_i, G_i), \forall i \in \mathbb{K}$, of system (4)–(5) are treated separately. The affine part is used in (15) to calculate $\lambda(x_e) \in \Lambda$ corresponding to the equilibrium point $x_e \in X_e$ of interest. Afterwards, the linear part represented by the linear matrix inequalities (LMIs) (14) together with (16) are equivalent to make $\sum_{i \in \mathbb{K}} \lambda_i(x_e) \mathcal{N}_i(\rho, P) < 0$. This inequality is a convex combination of N quadratic matrix functions for which $\mathcal{N}_\lambda(\rho, P) \leq \sum_{i \in \mathbb{K}} \lambda_i \mathcal{N}_i(\rho, P) < 0, \forall \lambda \in \Lambda$ and, consequently, a less conservative condition would be $\mathcal{N}_{\lambda(x_e)}(\rho, P) < 0$. In other words, we would like to obtain $\mathcal{H}_\infty$ design conditions based on a convex combination of the subsystems state-space matrices. Notice that for the subclass of switched affine systems with $(H_i, E_i, G_i) = (H, E, G), \forall i \in \mathbb{K}$, this goal is accomplished since we have $\mathcal{N}_{\lambda(x_e)}(\rho, P) = \sum_{i \in \mathbb{K}} \lambda_i(x_e) \mathcal{N}_i(\rho, P)$ for all $\lambda \in \Lambda$, see [7]. In this case, conditions (14) together with (16) are linear with respect to $\lambda \in \Lambda$ and are equivalent to

$$\begin{aligned} \mathcal{N}_{\lambda(x_e)}(\rho, P) &= A'_{\lambda(x_e)}P + PA_{\lambda(x_e)} + E'E \\ &+ (H'P + G'E)'(\rho I - G'G)^{-1}(H'P + G'E) < 0 \end{aligned} \quad (21)$$

which holds for some $P > 0$ if and only if

$$\|F(s)\|_\infty^2 < \rho \quad (22)$$

where $F(s) = E(sI - A_{\lambda(x_e)})^{-1}H + G$ is the transfer function that represents the system (4)–(5). Moreover, the associated switching function is

$$\sigma(x) = \arg \min_{i \in \mathbb{K}} (x - x_e)'P(A_i x + b_i) \quad (23)$$

As a conclusion, if the input matrix H_σ or the output ones (E_σ, G_σ) are switching dependent, it is not possible to obtain $\mathcal{H}_\infty$ design conditions based on convex combinations of the subsystems state-space matrices. The next theorem generalises the results to cope with state feedback control design.

Theorem 2: Consider system (1)–(2) and let $x_e \in \mathbb{R}^{n_x}$ be given. If there exist $\lambda \in \Lambda$, a positive definite matrix $S \in \mathbb{R}^{n_x \times n_x}$, symmetric matrices $R_i \in \mathbb{R}^{n_x \times n_x}$, matrices $Y_i \in \mathbb{R}^{n_u \times n_x}$ and a scalar $\rho > 0$

satisfying (15), $R_\lambda < 0$ and the conditions

$$\begin{bmatrix} A_i S + B_i Y_i + S A_i' + Y_i' B_i' - R_i & \bullet & \bullet \\ H_i' & -\rho I & \bullet \\ E_i S + F_i Y_i & G_i & -I \end{bmatrix} < 0, \quad i \in \mathbb{K} \quad (24)$$

then the control law (3) with $K_i = Y_i S^{-1}$ together with the state dependent switching strategy

$$\sigma(x) = \arg \min_{i \in \mathbb{K}} (x - x_e)' (S^{-1} R_i S^{-1} (x - x_e) + 2S^{-1} \ell_i) \quad (25)$$

make the equilibrium point $x_e \in \mathbb{R}^{n_x}$ globally asymptotically stable and assure that (6) holds.

Proof: The proof is direct. Multiplying (24) both sides by $\text{diag}(S^{-1}, I, I)$ with $S^{-1} = P$ and $Q_i = S^{-1} R_i S^{-1}$, we obtain the conditions of Theorem 1 with the replacements $(A_i + B_i K_i) \rightarrow A_i$ and $(E_i + F_i K_i) \rightarrow E_i$. $\square$

Notice that to obtain conditions based on a convex combination of the subsystems state-space matrices, we need to include the additional constraints $K_i = K, \forall i \in \mathbb{K}$, and to consider a subclass of systems that presents $(H_i, E_i, F_i, G_i) = (H, E, F, G), \forall i \in \mathbb{K}$. In order to consider a wider subclass of switched affine system with this property, we need to design a more general switching function which is also dependent on the external input.

3.2 State-input dependent switching function

The next theorem provides conditions to the stabilisation problem of system (1)–(2) with $u(t) = 0, \forall t \geq 0$, and presents a state-input dependent switching function assuring that inequality (6) is verified for $\rho > 0$ given.

Theorem 3: Consider system (1)–(2) with $u(t) = 0, \forall t \geq 0$, and let $x_e \in \mathbb{R}^{n_x}$ be given. If there exist $\lambda \in \Lambda$, a positive definite matrix $P \in \mathbb{R}^{n_x \times n_x}$ and a scalar $\rho > 0$ satisfying (15) and the condition

$$\sum_{i \in \mathbb{K}} \lambda_i \mathcal{L}_i(\rho, P) < 0 \quad (26)$$

then the state-input dependent switching strategy

$$\sigma(x, w) = \arg \min_{i \in \mathbb{K}} \begin{bmatrix} x - x_e \\ w \end{bmatrix}' \mathcal{L}_i(\rho, P) \begin{bmatrix} x - x_e \\ w \end{bmatrix} + 2(x - x_e)' P \ell_i \quad (27)$$

makes the equilibrium point $x_e \in \mathbb{R}^{n_x}$ globally asymptotically stable and assures that (6) holds.

Proof: Consider that conditions (15) and (26) hold and adopt the switching function (27). The time derivative of (11) along an arbitrary trajectory of (4)–(5) provides

$$\begin{aligned} \dot{v}(\xi) &= \begin{bmatrix} \xi \\ w \end{bmatrix}' \mathcal{L}_\sigma(\rho, P) \begin{bmatrix} \xi \\ w \end{bmatrix} + 2\xi' P \ell_\sigma - z_e' z_e + \rho w' w \\ &= \min_{i \in \mathbb{K}} \begin{bmatrix} \xi \\ w \end{bmatrix}' \mathcal{L}_i(\rho, P) \begin{bmatrix} \xi \\ w \end{bmatrix} + 2\xi' P \ell_i - z_e' z_e + \rho w' w \\ &= \min_{\lambda \in \Lambda} \begin{bmatrix} \xi \\ w \end{bmatrix}' \sum_{i \in \mathbb{K}} \lambda_i \mathcal{L}_i(\rho, P) \begin{bmatrix} \xi \\ w \end{bmatrix} + 2\xi' P \ell_\lambda - z_e' z_e + \rho w' w \\ &< -z_e' z_e + \rho w' w \end{aligned} \quad (28)$$

where the second and third equalities come from the adoption of the switching function (27) and the inequality comes from (15) and (26). Finally, proceeding as in the proof of Theorem 1, the inequality $J_\infty(\sigma) < \rho$ is verified and the proof is concluded. $\square$

The manner to solve the conditions of Theorem 3 is the same adopted in Theorem 1. Condition (15) defines the set of attainable equilibrium points (20). After, choosing the equilibrium point of interest, condition (26) provides $P(\lambda(x_e))$ and $\rho(\lambda(x_e))$ that are important to implement the switching function (27). Notice that the following inequalities $\mathcal{L}_\lambda(\rho, P) \leq \sum_{i \in \mathbb{K}} \lambda_i \mathcal{L}_i(\rho, P) < 0$ put in evidence that a less conservative condition is that one based on a convex combination of the subsystems state-space matrices. This condition can be obtained by assuming that $(E_i, G_i) = (E, G)$, $i \in \mathbb{K}$, since in this case, we have $\mathcal{L}_{\lambda(x_e)}(\rho, P) = \sum_{i \in \mathbb{K}} \lambda_i(x_e) \mathcal{L}_i(\rho, P)$. Moreover, for this subclass of switched systems the feasibility of $\mathcal{L}_{\lambda(x_e)}(\rho, P) < 0$ is verified if and only if (22) with $F(s) = E(sI - A_{\lambda(x_e)})^{-1}H_{\lambda(x_e)} + G$ holds and the switching function becomes linear with respect to the pair (A_i, H_i)

$$\sigma(x, w) = \arg \min_{i \in \mathbb{K}} \begin{bmatrix} x - x_e \\ w \end{bmatrix}' \begin{bmatrix} A_i'P + PA_i & \bullet \\ H_i'P & -\rho I \end{bmatrix} \begin{bmatrix} x - x_e \\ w \end{bmatrix} + 2\xi'P\ell_i \quad (29)$$

Notice that, for both switching rules, whenever $x_e \in X_e$ is chosen, the associated vector $\lambda(x_e) \in \Lambda$ is obtained, and the conditions are expressed in terms of LMIs, being solved without difficulty by any software available in the literature [26]. The next theorem generalises the conditions to cope with state feedback control design.

Theorem 4: Consider system (1)–(2) and let $x_e \in \mathbb{R}^{n_x}$ be given. If there exist $\lambda \in \Lambda$, a positive definite matrix $S \in \mathbb{R}^{n_x \times n_x}$, symmetric matrices $R_i \in \mathbb{R}^{n_x \times n_x}$, $Z_i \in \mathbb{R}^{n_w \times n_w}$, matrices $Y_i \in \mathbb{R}^{n_u \times n_x}$, $J_i \in \mathbb{R}^{n_s \times n_w}$ and a scalar $\rho > 0$ satisfying (15), and the conditions

$$\begin{bmatrix} A_iS + B_iY_i + SA_i' + Y_i'B_i' - R_i & \bullet & \bullet \\ H_i' - J_i' & -\rho I - Z_i & \bullet \\ E_iS + F_iY_i & G_i & -I \end{bmatrix} < 0, \quad i \in \mathbb{K} \quad (30)$$

$$\begin{bmatrix} R_\lambda & \bullet \\ J_\lambda' & Z_\lambda \end{bmatrix} < 0, \quad i \in \mathbb{K} \quad (31)$$

then the control law (3) with $K_i = Y_iS^{-1}$ together with the state-input dependent switching strategy

$$\sigma(x, w) = \arg \min_{i \in \mathbb{K}} \begin{bmatrix} x - x_e \\ w \end{bmatrix}' \begin{bmatrix} S^{-1}R_iS^{-1} & \bullet \\ J_i'S^{-1} & Z_i \end{bmatrix} \begin{bmatrix} x - x_e \\ w \end{bmatrix} + 2(x - x_e)'P\ell_i \quad (32)$$

make the equilibrium point $x_e \in \mathbb{R}^{n_x}$ globally asymptotically stable and assure that (6) holds.

Proof: The proof is simple and will be presented in general lines. Performing the Schur complement to (30) with respect to the last row and column and multiplying the result both sides by $\text{diag}(S^{-1}, I)$ with $S^{-1} = P$, we obtain

$$\mathcal{L}_i(\rho, P) = \begin{bmatrix} S^{-1}R_iS^{-1} & \bullet \\ J_i'S^{-1} & Z_i \end{bmatrix} < 0, \quad i \in \mathbb{K} \quad (33)$$

making the replacements $(A_i + B_iK_i) \rightarrow A_i$ and $(E_i + F_iK_i) \rightarrow E_i$ in $\mathcal{L}_i(\cdot)$. Inequality (33) together with (31) assures that $\mathcal{L}_{\lambda(x_e)}(\cdot) < 0$ for $\lambda(x_e)$ obtained from (15). The switching strategy follows from (33) using the same arguments as in Theorem 1. The proof is concluded. $\square$

In this case, matrices $(E_i, F_i, G_i) = (E, F, G)$ for all $i \in \mathbb{K}$ and the additional restrictions $K_i = K$, $\forall i \in \mathbb{K}$, must be imposed in order to obtain conditions based on a convex combination of the subsystems state-space matrices. An important remark is that, in this case, the set of attainable equilibrium points (20) does not need

that matrix A_λ be Hurwitz. Indeed, because of the additional control variable (3) the set $\mathcal{H}$ is composed by all non-singular matrices A_λ , $\lambda \in \Lambda$, such that $\sum_{\lambda \in \Lambda} \lambda_i(A_i + B_iK_i)$ is Hurwitz with K_i , $i \in \mathbb{K}$, to be determined, see inequalities (33) together with (31). The conditions presented in this paper assure global asymptotic stability and $\mathcal{H}_\infty$ performance even in the eventual occurrence of sliding modes.

3.3 Comparison with max-type Lyapunov function-based approach

The literature to date presents recent works that propose stability conditions for switched affine systems based on a max-type Lyapunov function, as for instance, [21–24], where the last reference treats $\mathcal{H}_\infty$ control design as well.

In [24], a necessary condition for $\mathcal{H}_\infty$ performance is the following (see inequality (42) of [24] for $\theta = \bar{\theta}$)

$$\begin{bmatrix} A_\lambda'P_\lambda + P_\lambda A_\lambda & \bullet \\ H_\lambda'P_\lambda & -\rho I \end{bmatrix} + \begin{bmatrix} E_\lambda' \\ G_\lambda' \end{bmatrix} \begin{bmatrix} E_\lambda' \\ G_\lambda' \end{bmatrix}' < 0 \quad (34)$$

for a particular $\lambda(x_e) \in \Lambda$ satisfying (15). The inequality (34) is expressed with the notation defined in this paper. Notice that, for $(E_i, G_i) = (E, G)$, $\forall i \in \mathbb{K}$, condition (34) is equivalent to (26), which is sufficient to assure an $\mathcal{H}_\infty$ performance cost according to Theorem 3. Actually, the performance conditions proposed in [24] (see inequality (42) of [24]) must be imposed for all $\lambda \in \Lambda$. On the contrary, inequality (34) expresses only one of the restrictions for a particular $\lambda(x_e) \in \Lambda$. Hence, under switching-independent output matrices the conditions of Theorem 3 are less conservative than the ones proposed in [24]. The same conclusion can be drawn for $(H_i, E_i, G_i) = (H, E, G)$, $\forall i \in \mathbb{K}$ and adopting Theorem 1. Moreover, as it will be clear in Example 1, even for switching-dependent input matrices H_σ , the simpler state-dependent switching rule proposed in Theorem 1 provides a performance cost much smaller than the one proposed in [24]. This indicates that the set of restrictions proposed here is softer than the ones required by adopting a max-type Lyapunov function approach.

3.4 Comparison with practical stability

Some recent works, as for instance [19], provide conditions that assure practical stabilisation of switched affine systems. In [19], two different switching rules are proposed based on a quadratic Lyapunov function and switched Lyapunov function, respectively. Both are functions of $\xi_k = \xi(t_k)$ which is assumed constant in the time interval $t \in T_k = t_{k+1} - t_k \leq T_{\max}$. These rules assure that the system trajectories are exponentially attracted to the ellipsoid

$$\mathcal{E}^{T_{\max}} = \left\{ \xi \in \mathbb{R}^{n_x} : \bar{v}(\xi) \leq \frac{\beta T_{\max}}{2\gamma} \right\} \quad (35)$$

whenever $t \geq 0$ goes to infinity, where $\bar{v}(\xi) = \xi'P\xi$ in the quadratic case and $\bar{v}(\xi) = \min_{j \in \mathcal{J}} \text{eig}_{\min}(P_j)|\xi|^2 \leq (\beta T_{\max})/(2\gamma)$ in the switched one. In order to compare with our results, we need to impose $T_{\max} = 0$ and $\gamma \rightarrow 0$. With these choices, [19] provides

$$A_\lambda'P + PA_\lambda < 0 \quad (36)$$

for the quadratic case (see Theorem 1 of [19]) and

$$A_{\lambda^j}'P_j + P_jA_{\lambda^j} + \sum_{r=1}^M \mu_{jr}(P_r - P_j) < 0, \quad j \in \mathcal{J} \quad (37)$$

for the switched one, where the set $\mathcal{J} = \{1, \dots, M\}$ is composed by all vectors $\lambda^j \in \Lambda$ such that $\ell_\lambda = 0$ (see Theorem 2 of [19]). However, in most of practical applications (see Example 1 of this paper) the set $\mathcal{J}$ is unitary, that means, for each x_e there exists only one $\lambda(x_e) \in \Lambda$ satisfying $\ell_\lambda = 0$. In this case, condition (37) reduces to

(36) leading to the conclusion that, as presented in [3, 18], a sufficient condition for global asymptotic stability is that matrix $A_{\lambda(x_e)}$ be Hurwitz. Moreover, it seems to be a very difficult task to generalise the conditions proposed in [19] to cope with state feedback control design. The next section presents two examples solved in an Apple computer with operational system OS X version 10.9.3 and using Matlab R2013b facilities.

4 Illustrative examples

4.1 Example 1

This example was borrowed from [24] and consists of a buck-boost converter described by the state-space equations (1) and (2) with $u(t) = 0$ and matrices

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \quad (38)$$

$$H_1 = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, \quad H_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad b_1 = \begin{bmatrix} \frac{E_{in}}{L} \\ 0 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (39)$$

$E_1 = E_2 = [0 \ 1]$ and $G_1 = G_2 = 0$ where $E_{in} = 15 \text{ V}$, $R = 30 \ \Omega$, $L = 10^{-3} \text{ H}$ and $C = 10^{-6} \text{ F}$. To make the problem more amenable for numerical purposes we have scaled the matrices $\{A_i, H_i, b_i\}$ by $\{10^{-3}A_i, 10^{-3}H_i, 10^{-3}b_i\}$ in order to consider ms – time basis instead of sec – time basis. We have selected the same equilibrium point adopted in [24] for the converter operating as a buck, that is, $x_e = [0.48 \ -9] \in X_e$ associated with $\lambda(x_e) = [0.375 \ 0.625] \in \Lambda$ and we have solved the conditions of Theorems 1 and 3. Applying the external disturbance $w(t) = \sin(2\pi 0.12t)$, $8.3 \leq t < 33.3$ and $w(t) = 0$ otherwise, we have implemented the switching strategies, obtaining the state trajectories presented in Fig. 1. In both cases, they are almost coincident and the trajectories rapidly evolve on a stable sliding mode.

In order to consider the case where the converter operates as a boost, we have chosen the same equilibrium point proposed in [24], that is, $x_e = [1.68 \ -21] \in X_e$ associated with $\lambda(x_e) = [0.583 \ 0.417] \in \Lambda$. Table 1 presents the $\mathcal{H}_\infty$ guaranteed cost ρ for the two equilibrium points, considering the conditions of Theorems 1, 3 and condition of [24], as well as, the actual costs $J_\infty^{\sigma_x}$ and $J_\infty^{\sigma_{x,w}}$ for the external input $w(t)$ already defined. Notice that, the costs obtained by the strategies proposed in this paper are much smaller than those based on the max-type Lyapunov function. Moreover, as the pair $(E_i, G_i) = (E, G)$, $\forall i \in \mathbb{K}$ is switching independent, the $\mathcal{H}_\infty$ guaranteed cost provided by Theorem 3 equals $\|E(sI - A_\lambda)^{-1}H_\lambda + G\|_\infty^2$ as expected.

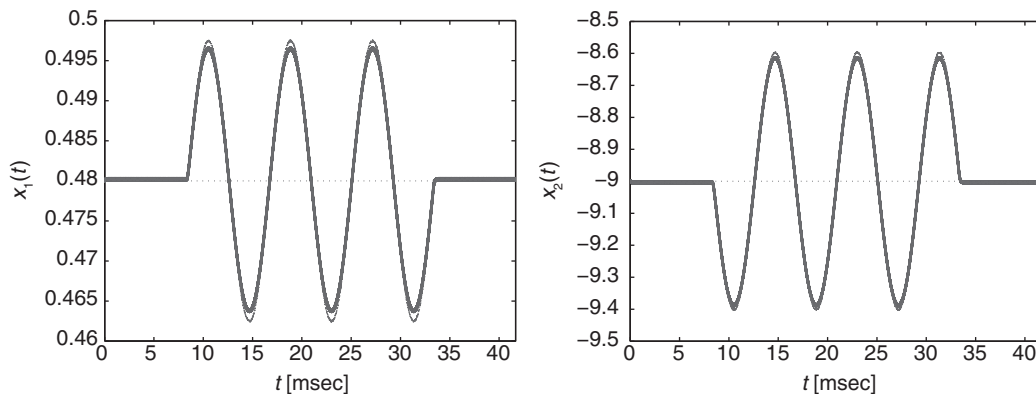


Fig. 1 Buck-boost converter state trajectories for $x_e = [0.48 \ -9]'$

Table 1 Performance comparison

x_e	$\mathcal{H}_\infty$ cost	Ref. [24]	Theorem 1	Theorem 3
[0.48 -9]'	ρ	23.983	0.9600	0.3600
	$J_\infty^{\sigma_x}$	–	0.1613	–
	$J_\infty^{\sigma_{x,w}}$	–	–	0.1475
[1.68 -21]'	ρ	15.301	3.3599	1.9599
	$J_\infty^{\sigma_x}$	–	1.7051	–
	$J_\infty^{\sigma_{x,w}}$	–	–	1.6334

4.2 Example 2

This example was inspired on [27] and consists of system (1)–(2) defined by matrices

$$A_1 = \begin{bmatrix} -5 & 0 \\ 0 & 8 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 4 \end{bmatrix} \quad (40)$$

$$H_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad H_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \quad F = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (41)$$

where $B_1 = B_2 = B$, $E_1 = E_2 = I$, $F_1 = F_2 = F$ and $G_1 = G_2 = [1 \ 1]'$. Notice that both matrices A_1 and A_2 are unstable and there is no Hurwitz convex combination A_λ . Consequently, it is not possible to use the conditions of Theorem 1 nor Theorem 3, neither the ones proposed in [19, 24] to assure global asymptotic stability. Moreover, the pairs (A_1, B_1) and (A_2, B_2) are not controllable. We have obtained the set of equilibrium points (20) and chosen the point $x_e = [0.5 \ -1.5]'$ correspondent to $\lambda = [0.5 \ 0.5]'$ to be attained. Solving the conditions of Theorem 2 we have obtained $\rho = 5.4005$ and the matrix gains

$$K_1 = [0.8328 \ -5.3264], \quad K_2 = [0.9857 \ -5.1345] \quad (42)$$

On the other hand, solving the conditions of Theorem 4, we have obtained $\rho = 5.2900$ and

$$K_1 = [0.7943 \ -4.8723], \quad K_2 = [0.7754 \ -4.7977] \quad (43)$$

Using the results of Theorem 4, Fig. 2 presents the state trajectories and the control law (3) for an external input $w(t) = te^{-0.5t}$. The actual cost is $J_\infty^{\sigma(x,w)} = 2.0609$ which is much smaller than the obtained guaranteed cost. This indicates that the external input is not too severe. This example shows the importance of the joint design of two control variables, the switching function and the control law (3), in order to assure stability and $\mathcal{H}_\infty$ performance of switched affine systems.

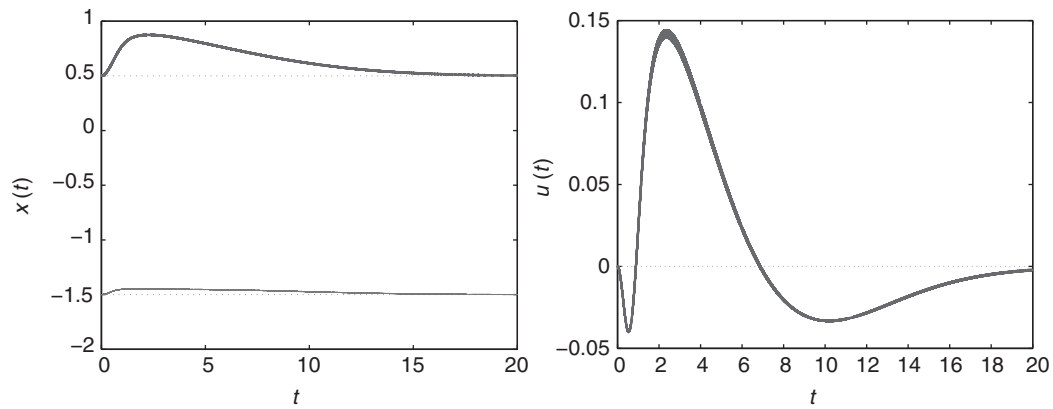


Fig.2 State trajectories and control law

5 Conclusion

This paper has proposed two different switching strategies to cope with $\mathcal{H}_\infty$ control design of continuous-time switched affine systems. The first one is state dependent and the other, more general, depends also on the external input. As a first step, conditions were obtained in order to design a switching function as the unique control variable. Afterwards, these conditions were generalised to cope with a state feedback control design problem characterised by the joint design of a switching rule and a set of state feedback gains. A theoretical comparison has shown that both rules are based on less conservative conditions than others from the literature, in particular, those obtained from the max-type Lyapunov function approach. Moreover, contrary to the conditions for practical stability, they allow simple generalisation to deal with more general control design problems. Two illustrative examples showed the efficiency and the validity of the proposed theory.

6 Acknowledgements

This research was supported by grants from ‘Fundação de Amparo à Pesquisa do Estado de São Paulo. – FAPESP’, Brazil.

7 References

- Cardim, R., Teixeira, M.C.M., Assunção, E., Covacic, M.A.: ‘Variable-structure control design of switched systems with an application to a DC-DC power converter’, *IEEE Trans. Ind. Electron.*, 2009, **56**, pp. 3505–3513
- Corona, D., Buisson, J., De Schutter, B., Giua, A.: ‘Stabilization of switched affine systems: an application to the buck–boost converter’. Proc. of the 2007 American Control Conf., New York City, USA, 2007, pp. 6037–6042
- Deaecto, G.S., Geromel, J.C., Garcia, F.S., Pomilio, J.A.: ‘Switched affine systems control design to DC–DC converters’, *IET Control Theory Appl.*, 2010, **4**, pp. 1201–1210
- Mahmoud, M.S., Shi, P., Saif, A.W.A.: ‘Stabilization of linear switched delay systems: $\mathcal{H}_2$ and $\mathcal{H}_\infty$ methods’, *J. Optim. Theory Appl.*, 2009, **142**, pp. 583–601
- Wu, L., Zheng, W.X., Gao, H.: ‘Dissipativity-based sliding mode control of switched stochastic systems’, *IEEE Trans. Autom. Control*, 2013, **58**, pp. 785–791
- Geromel, J.C., Colaneri, P.: ‘Stability and stabilization of continuous-time switched linear systems’, *SIAM J. Control Optim.*, 2006, **45**, pp. 1915–1930
- Zhai, G.: ‘Quadratic stabilizability and $\mathcal{H}_\infty$ disturbance attenuation of switched linear systems via state and output feedback’. Proc. of the 51st IEEE Conf. on Decision and Control, Maui, USA, 2012, pp. 1935–1940
- Deaecto, G.S., Geromel, J.C., Daafouz, J.: ‘Dynamic output feedback $\mathcal{H}_\infty$ control of switched linear systems’, *Automatica*, 2011, **47**, pp. 1713–1720
- Geromel, J.C., Colaneri, P., Bolzern, P.: ‘Dynamic output feedback control of switched linear systems’, *IEEE Trans. Autom. Control*, 2008, **53**, pp. 720–733
- Geromel, J.C., Deaecto, G.S., Daafouz, J.: ‘Suboptimal switching control consistency analysis for switched linear systems’, *IEEE Trans. Autom. Control*, 2013, **58**, pp. 1857–1861
- DeCarlo, R.A., Branicky, M.S., Pettersson, S., Lennartson, B.: ‘Perspectives and results on the stability and stabilizability of hybrid systems’, *Proc. IEEE*, 2000, **88**, pp. 1069–1082
- Lin, H., Antsaklis, P.J.: ‘Stability and stabilizability of switched linear systems: a survey of recent results’, *IEEE Trans. Autom. Control*, 2009, **54**, pp. 308–322
- Shorten, R., Wirth, F., Mason, O., Wulff, K., King, C.: ‘Stability criteria for switched and hybrid systems’, *SIAM Rev.*, 2007, **49**, pp. 545–592
- Liberzon, D.: ‘Switching in Systems and Control’ (Birkhäuser, Boston, USA, 2003)
- Sun, Z., Ge, S.S.: ‘Switched linear systems: control and design’ (Springer, London, 2005)
- Hauroigne, P., Riedinger, P., Jung, C.: ‘Switched affine systems using sampled-data controllers: robust and guaranteed stabilization’, *IEEE Trans. Autom. Control*, 2011, **56**, pp. 2929–2935
- Seatzu, C., Corona, D., Giua, A., Bemporad, A.: ‘Optimal control of continuous-time switched affine systems’, *IEEE Trans. Autom. Control*, 2006, **51**, pp. 726–741
- Bolzern, P., Spinelli, W.: ‘Quadratic stabilization of a switched affine system about a nonequilibrium point’. Proc. of the 2004 American Control Conf., Boston, Massachusetts, 2004, pp. 3890–3895
- Hetel, L., Fridman, E.: ‘Robust sampled-data control of switched affine systems’, *IEEE Trans. Autom. Control*, 2013, **58**, pp. 2922–2928
- Kuiava, R., Ramos, R.A., Pota, H.R., Alberto, L.F.C.: ‘Practical stability of switched systems without a common equilibria and governed by a time-dependent switching signal’, *Eur. J. Control*, 2013, **19**, pp. 206–213
- Trofino, A., Assmann, D., Scharlau, C.C., Coutinho, D.F.: ‘Switching rule design for switched dynamic systems with affine vector fields’, *IEEE Trans. Autom. Control*, 2009, **54**, pp. 2215–2222
- Trofino, A., Assmann, D., Scharlau, C.C., Coutinho, D.F.: ‘Corrections to Switching rule design for switched dynamic systems with affine vector fields’, *IEEE Trans. Autom. Control*, 2012, **57**, pp. 1080–1082
- Trofino, A., Scharlau, C.C., Dezu, T.J.M., de Oliveira, M.C.: ‘Stabilizing switching rule for affine switched systems’. Proc. of the 50th IEEE Conf. on Decision and Control, Florida, USA, 2011, pp. 1183–1188
- Trofino, A., Scharlau, C.C., Dezu, T.J.M., de Oliveira, M.C.: ‘Switching rule design for affine switched systems with $\mathcal{H}_\infty$ performance’. Proc. of the 51th IEEE Conf. on Decision and Control, Maui, USA, 2012, pp. 1923–1928
- Xu, X., Zhai, G., He, S.: ‘On practical asymptotic stabilizability of switched affine systems’, *Nonlinear Anal. Hybrid Syst.*, 2008, **2**, pp. 196–208
- Boyd, S.P., El Ghaoui, L., Feron, E., Balakrishnan, V.: ‘Linear matrix inequalities in system and control theory’ (SIAM, Philadelphia, 1994)
- Xie, G., Wang, L.: ‘Periodical stabilization of switched linear systems’, *J. Comput. Appl. Math.*, 2005, **181**, pp. 176–187

Copyright of IET Control Theory & Applications is the property of Institution of Engineering & Technology and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.